Optimal Sequence Alig-ment

Plan 9 from outer space
https://commons.wikimedia.org/w/index.php?curid=1793456

Overview

- The alignment problem
- The dynamic programming solution
- Pairwise alignment: exact global and local solutions
- Multiple alignment and the cost of perfection

Recap: protein scoring

$$\left(\frac{C_{a,b}}{f(a)f(b)} \right)$$

C matrix – scaled frequencies of change from amino acid a to amino acid b (based on observed changes in some set)

Expectation based solely on frequencies of amino acids (changes not favoured / disfavoured)

Better than random: ratio > 1
Random: ratio = 1
Worse than random: ratio < 1

log transformation

$$D_{a,b} = S \cdot \log\left(\frac{C_{a,b}}{f(a)f(b)}\right)$$

Magic

log

Better than random: $D_{a,b} > 0$
Random: $D_{a,b} = 0$
Worse than random: $D_{a,b} < 0$

PAM150 matrix (S = 2, log base 2)
Half-bits

	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W	
C	9																				C
S	-1	4																			S
T	-1	1	5																		T
P	-3	-1	-1	7																	P
A	0	1	0	-1	4																A
G	-3	0	-2	-2	0	6															G
N	-3	1	0	-2	-2	0	6														N
D	-3	0	-1	-1	-2	-1	1	6													D
E	-4	0	-1	-1	-1	-2	0	2	5												E
Q	-3	0	-1	-1	-1	-2	0	0	2	5											Q
H	-3	-1	-2	-2	-2	-2	1	-1	0	0	8										H
R	-3	-1	-1	-2	-1	-2	0	-2	0	1	0	5									R
K	-3	0	-1	-1	-1	-2	0	-1	1	1	-1	2	5								K
M	-1	-1	-1	-2	-1	-3	-2	-3	-2	0	-2	-1	-1	5							M
I	-1	-2	-1	-3	-1	-4	-3	-3	-3	-3	-3	-3	-3	1	4						I
L	-1	-2	-1	-3	-1	-4	-3	-4	-3	-2	-3	-2	-2	2	2	4					L
V	-1	-2	0	-2	0	-3	-3	-3	-2	-2	-3	-3	-2	1	3	1	4				V
F	-2	-2	-2	-4	-2	-3	-3	-3	-3	-3	-1	-3	-3	0	0	0	-1	6			F
Y	-2	-2	-2	-3	-2	-3	-2	-3	-2	-1	2	-2	-2	-1	-1	-1	-1	3	7		Y
W	-2	-3	-2	-4	-3	-2	-4	-4	-3	-2	-2	-3	-3	-1	-3	-2	-3	1	2	11	W
	C	S	T	P	A	G	N	D	E	Q	H	R	K	M	I	L	V	F	Y	W	

https://bioinformaticamente.com/2020/10/22/alignment-algorithms/

DNA matrix

Something like this usually works:

	A	G	C	T
A	1	-1	-1	-1
G	-1	1	-1	-1
C	-1	-1	1	-1
T	-1	-1	-1	1

Or this:

	A	G	C	T
A	1	0.5	-1	-1
G	0.5	1	-1	-1
C	-1	-1	1	0.5
T	-1	-1	0.5	1

Back to the alignment problem

Given a scoring scheme…

and a set of homologous sequences S…

introduce gaps if necessary to generate an alignment that optimizes the score

So let's make some alignments!

Sequence S_1: length m
Sequence S_2: length n

So let's make some alignments!

Sequence S_1: length m

Sequence S_2: length n

In total, there are $\binom{n+m}{m}$ possible alignments of these sequences

n = m = 2:

4!/2!2! = 6 possibilities

```
AB--   AB-   AB-   AB   A-B   -AB
--CD   -CD   C-D   CD   -CD   CD-
```

n = m = 10: 184,756 possible alignments

Alignment of 2 sequences, each 100 amino acids in length:

$= 9.05485147 \times 10^{58}$ possibilities

Brute force is *not* going to work here…

Scaling of algorithms: Big-O Notation

What rate do resources (time, memory) increase as the input increases?

Asymptotic: upper bound on growth as input tends to infinity

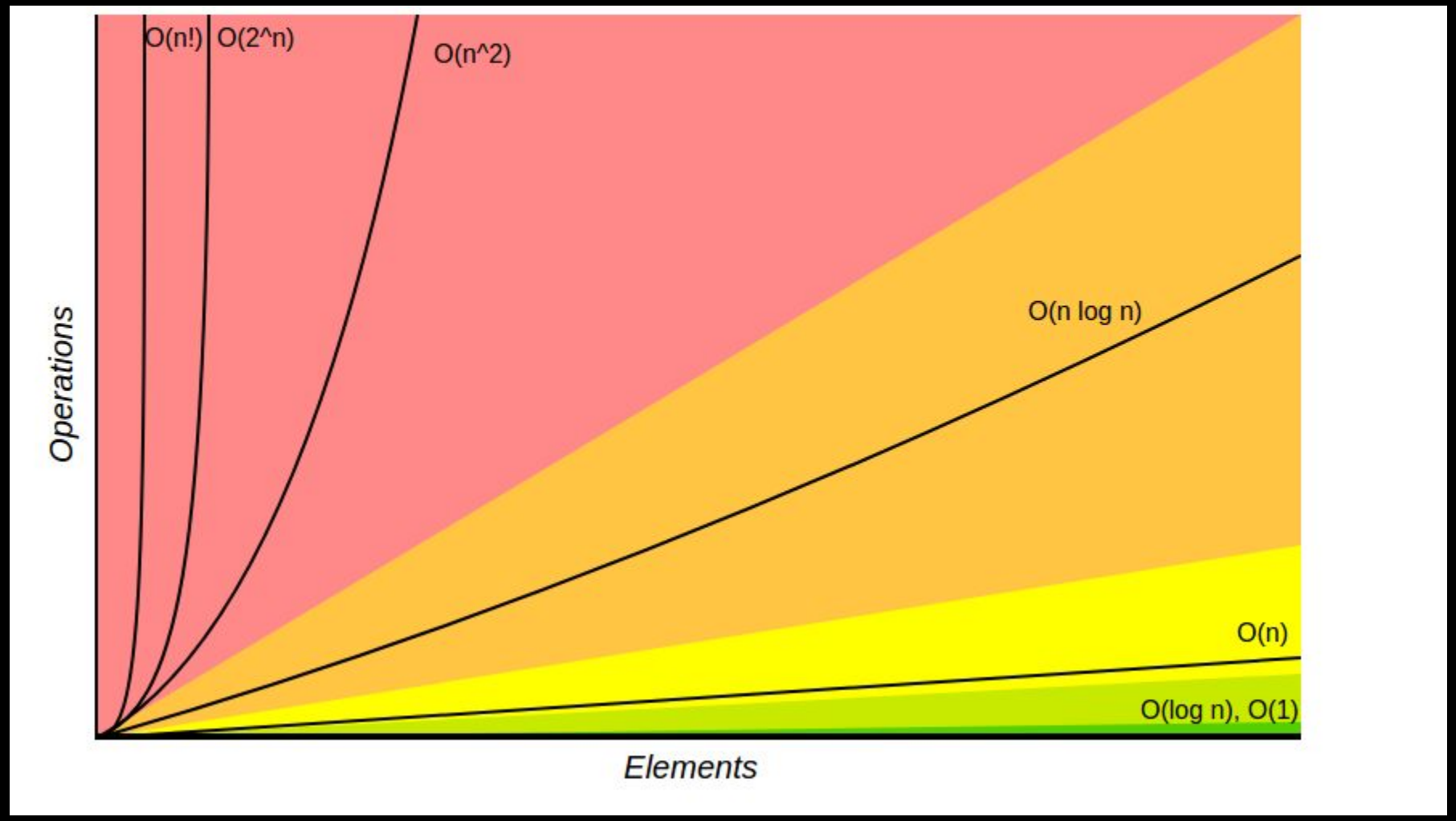
Keep only dominant term:

$3n^2 + 7n + 42$ is $O(n^2)$

Brute-force sequence alignment:

2n choose n $\sim$ $O(4^n)$

Big O complexity



The Key to Alignment

If we were given a **point X** within an optimal alignment of S_1 and S_2, we could **split on X** and solve each problem independently

```
MEH..KNP..TYL
MDH..KQP..SYI
```

```
MEH..K
MDH..K
```

+

```
P..TYL
P..SYI
```

But we **don't know** any X, so divide and conquer isn't going to work

However…

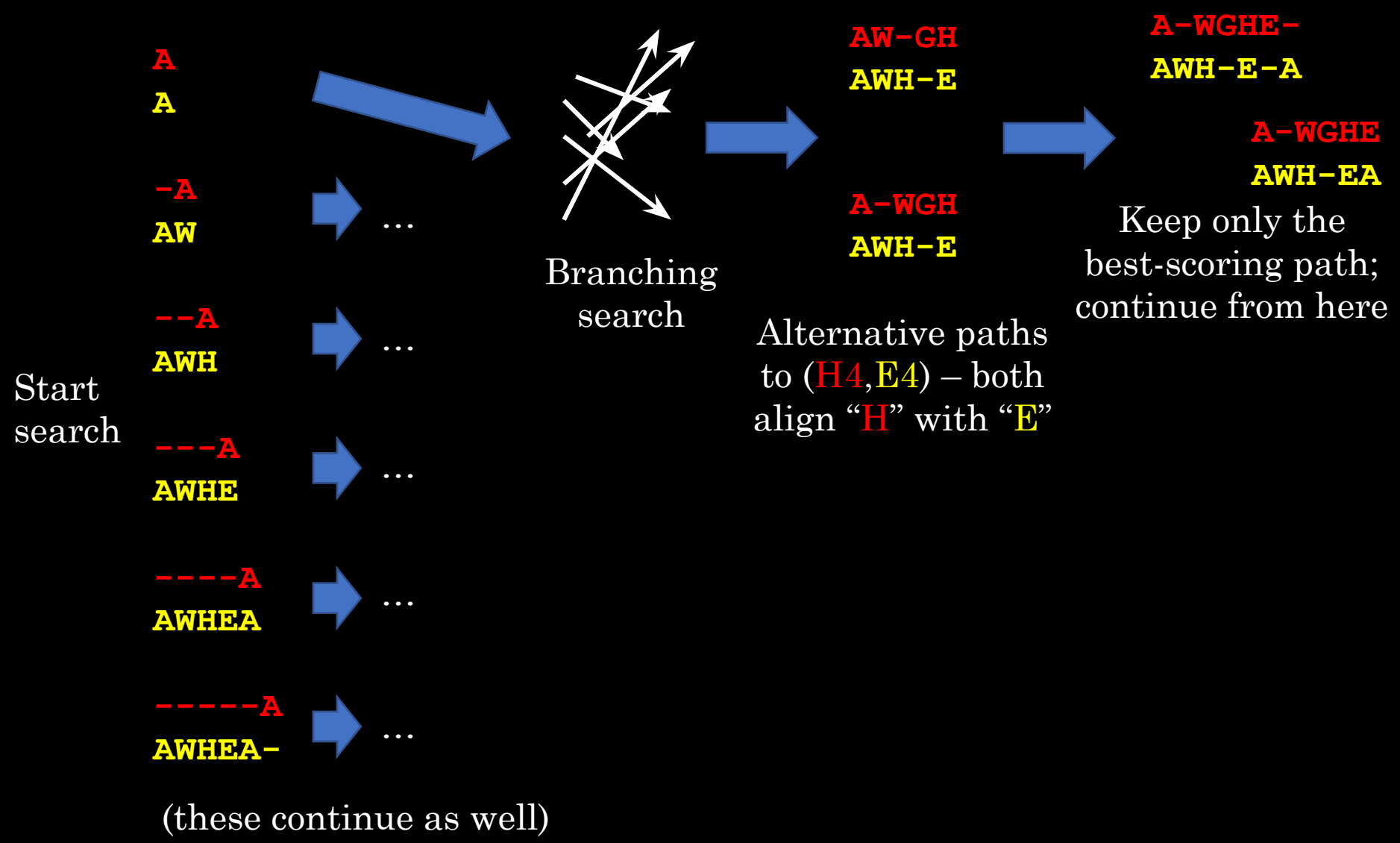
In searching for the best alignment:

- Start at the beginning of the sequences and consider every possible X

- BUT -

- Store only the best path (series of matches and gaps) that leads us to X

Consider an alignment search for AWGHE vs AWHEA:



= Dynamic Programming

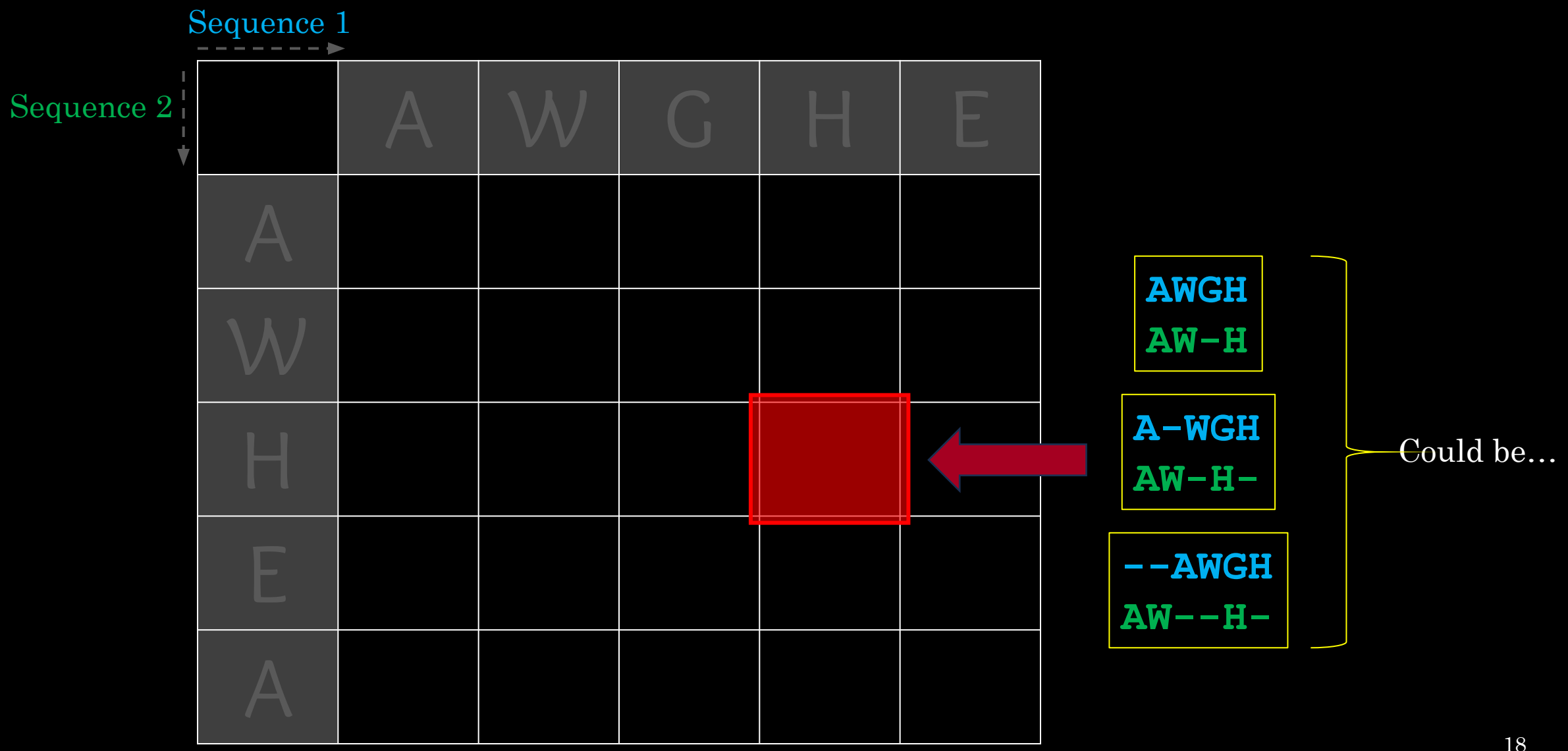
Consider an alignment of AWGHE vs AWHEA:

Sequence 1

Sequence 2

	A	W	G	H	E
A					
W					
H					
E					
A					

Each cell in the grid represents a point in the alignment where the corresponding residues have been added to the alignment



Determine the best score for every possible X from the two sequences

AWGHE vs. AWHEA	A	W	G	H	E
A	Best →(A,A)				
W					
H				Best → (H,H)	
E					
A					Best → (A,E)

Filling the matrix

We need our substitution matrix S and gap penalty scheme G

(we'll start with a linear gap penalty G = -gd)

For each possible X, consider the three immediate precursors

Upper left-hand corner: set to 0

S = **PAM250**
g = **5**

AWGHE vs. AWHEA		A	W	G	H	E
	0					
A						
W						
H						
E						
A						

First row / columns: runs of initial gaps

AWGHE
vs.
AWHEA

		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5					
W	-10					
H	-15					
E	-20					
A	-25					

AWGHE-----
-----AWHEA

S = **PAM250**
g = **5**

-----AWGHE
AWHEA-----

insert gap in AWGHE

insert gap in AWHEA

Filling (A,A): What is the best path to get there?

S = **PAM250**
g = **5**

AWGHE vs. AWHEA		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5					
W	-10					
H	-15					
E	-20					
A	-25					

insert gap in AWGHE

insert gap in AWHEA

match

−A	A
A	−A
A	A

Choosing the best path to (A,A)

S(A,A) = 2

Therefore:

Insert -10

Insert -10

Match 2

AWGHE vs. AWHEA		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5	2				
W	-10					
H	-15					
E	-20					
A	-25					

General form: best path to any matrix cell

AWGHE vs. AWHEA		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5					
W	-10					
H	-15					
E	-20					
A	-25					

F(2,2) F(2,3)

F(3,2) F(3,3) = ?

F(3,3) = max

- F(2,2) + S(G,H) — match
- F(2,3) - *d* — insert gap in AWGHE
- F(3,2) - *d* — insert gap in AWHEA

Remember paths INTO (not out of) each cell

AWGHE vs. AWHEA		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5	2	-3	-8	-13	-18
W	-10	-3	19	14	9	4
H	-15	-8	14	17	20	15
E	-20	-13	9	14	18	24
A	-25	-18	4	10	13	19

Global Exact Alignment: Needleman-Wunsch

Since we have retained the best path to each F(x,y) in the matrix, we can <u>trace back</u> from the endpoint F(m,n) to the origin and retrieve the optimal alignment path

Traceback for the optimal alignment

AWGHE
vs.
AWHEA

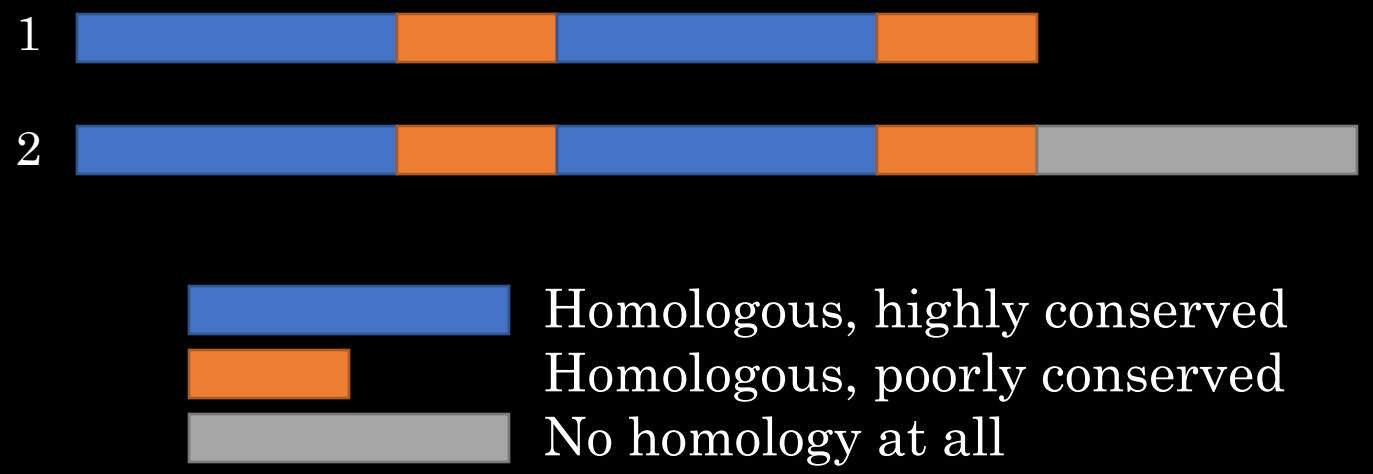
		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5	2	-3	-8	-13	-18
W	-10	-3	19	14	9	4
H	-15	-8	14	17	20	15
E	-20	-13	9	14	18	24
A	-25	-18	4	10	13	19

AWGHE-
AW-HEA

Local Exact Alignment: Smith-Waterman

- Only return 'good' sub-alignments of the whole problem

- Useful, for instance, when



One more rule for local alignment

AWGHE vs. AWHEA		A	W	G	H	E
	0	0	0	0	0	0
A	0					
W	0					
H	0					
E	0					
A	0					

F(2,2) F(2,3)

F(3,2) F(3,3) = ?

F(3,3) = max
- F(2,2) + S(G,H)
- F(2,3) - *d*
- F(3,2) - *d*
- **0**

match

insert gap in AWGHE

insert gap in AWHEA

Nothing is >0

The Needleman-Wunsch Matrix, Again

AWGHE-
AW-HEA

AWGHE vs. AWHEA		A	W	G	H	E
	0	-5	-10	-15	-20	-25
A	-5	2	-3	-8	-13	-18
W	-10	-3	19	14	9	4
H	-15	-8	14	17	20	15
E	-20	-13	9	14	18	24
A	-25	-18	4	10	13	19

The Smith-Waterman matrix

Slightly modified (non-trivial) S-W example

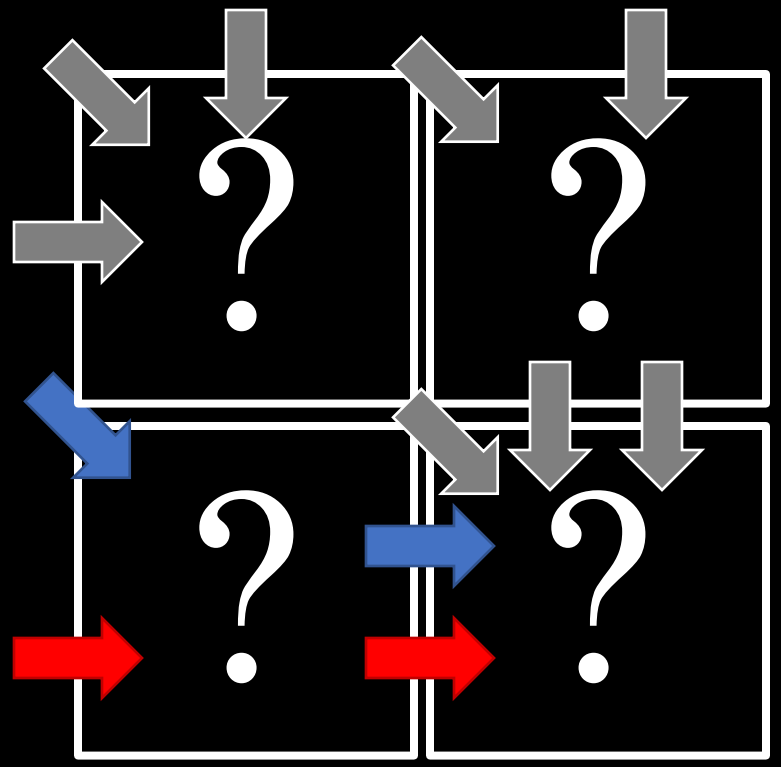
Find the **largest** value in the matrix, and trace back from there to 0

HE

HE

AWGHE vs. AYHEA		A	W	G	H	E
	0	0	0	0	0	0
A	0	2	0	1	0	0
Y	0	0	2	0	0	0
H	0	0	0	0	6	1
E	0	0	0	0	1	10
A	0	2	0	1	0	5

Affine Gap Penalties



Opening a new gap
(cost = d)

Extending a gap
(cost = e)

A horizontal move now has two possible costs; we need to consider both alternatives

(and therefore store the best scores for each box given horizontal, vertical, or diagonal entry)

Significance of S-W Alignments

Permutation test: Randomize the alignment n times, compute mean and standard deviation

Compute Z-score for each replicate:

Mean of replicates

True score

$$Z(A,B) = \frac{S(A,B) - \tilde{m}}{\tilde{\sigma}}$$

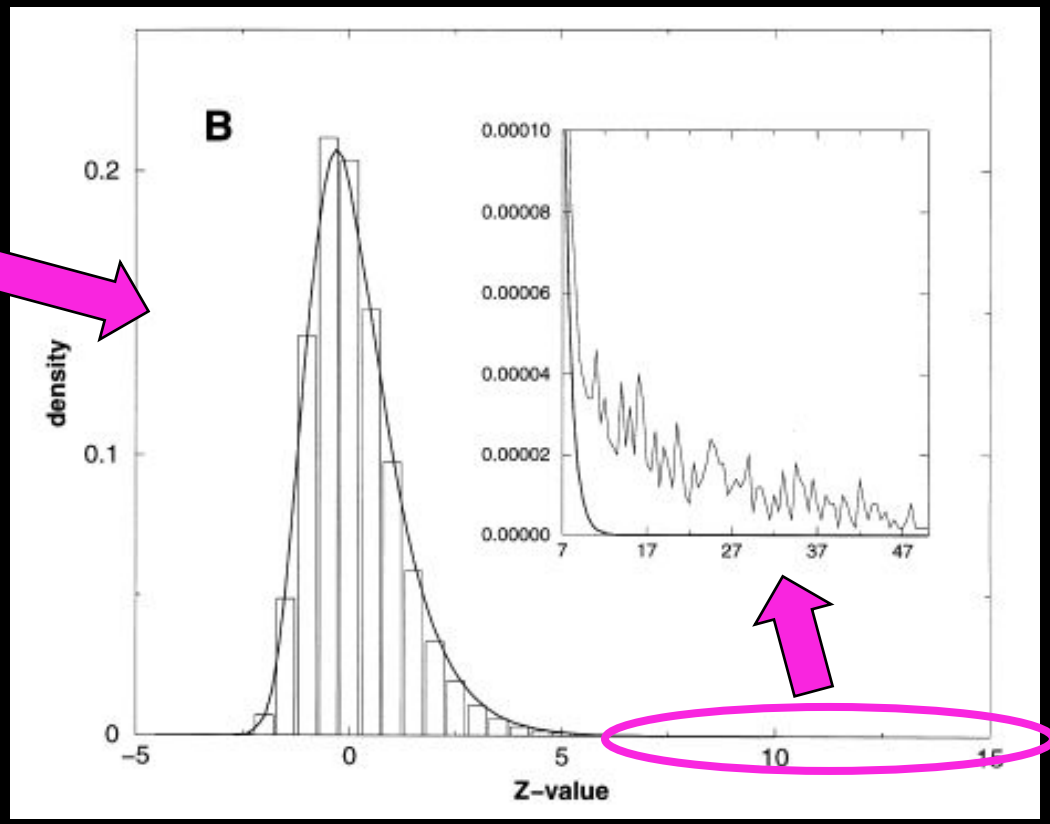
Standard deviation of replicates

Comet et al., *Computers and Chemistry* (1999)

Significance of S-W Alignments

$$Z(A,B) = \frac{S(A,B) - \tilde{m}}{\tilde{\sigma}}$$

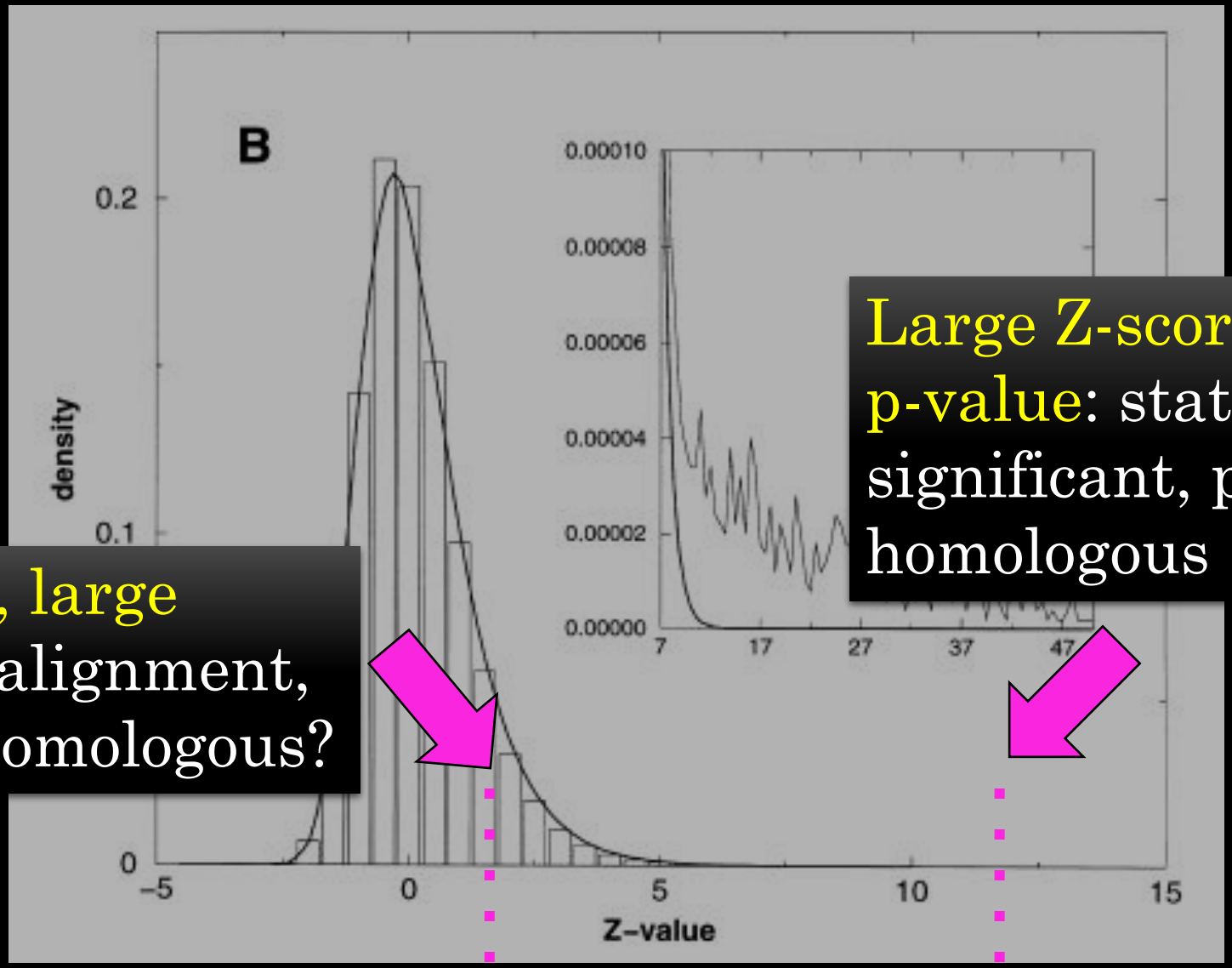
Curve = null model of Z-score fit to Gumbel extreme value distribution



Statistically "significant" alignments (small p-value)

Comet et al., *Computers and Chemistry* (1999)

Alignment Significance



Small Z-score, large p-value: poor alignment, possibly not homologous?

Large Z-score, small p-value: statistically significant, probably homologous

Comet et al., *Computers and Chemistry* (1999)

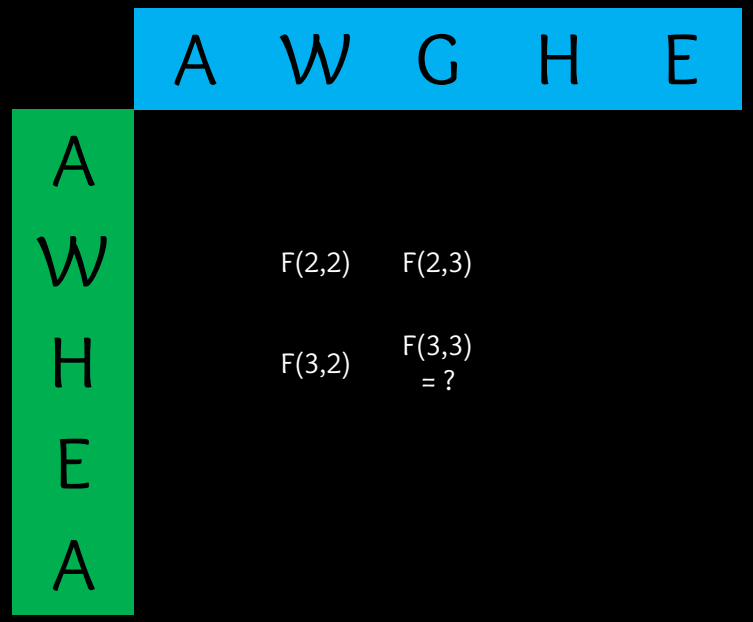
Alignment Complexity

- For each possible matching of a residue from sequence S_1 with a residue from S_2, we need to carry out a constant number of computations and comparisons

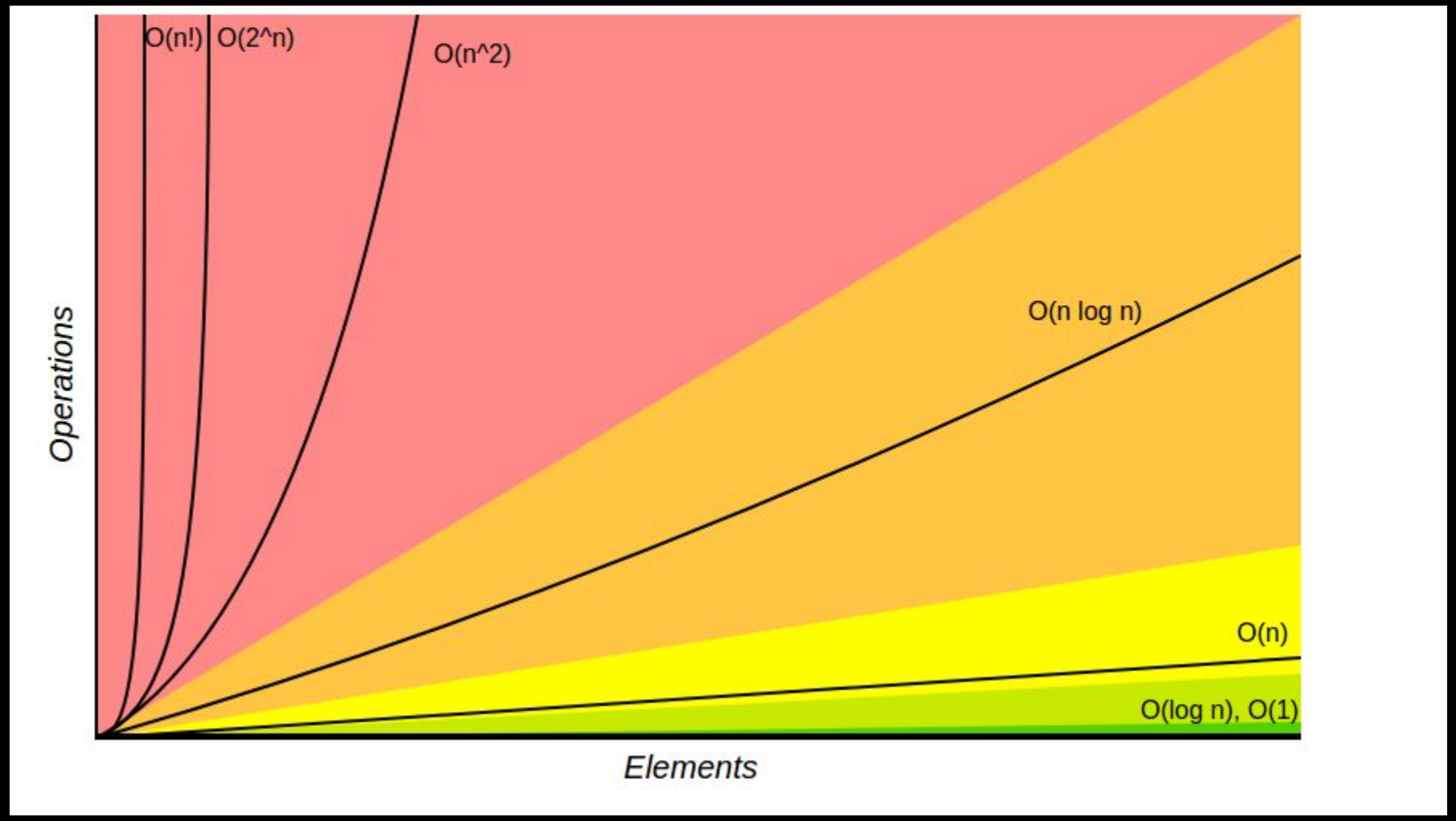
- Total = 3 x m x n = $O(mn)$

- ~ $O(n^2)$ if we assume $m \cong n$

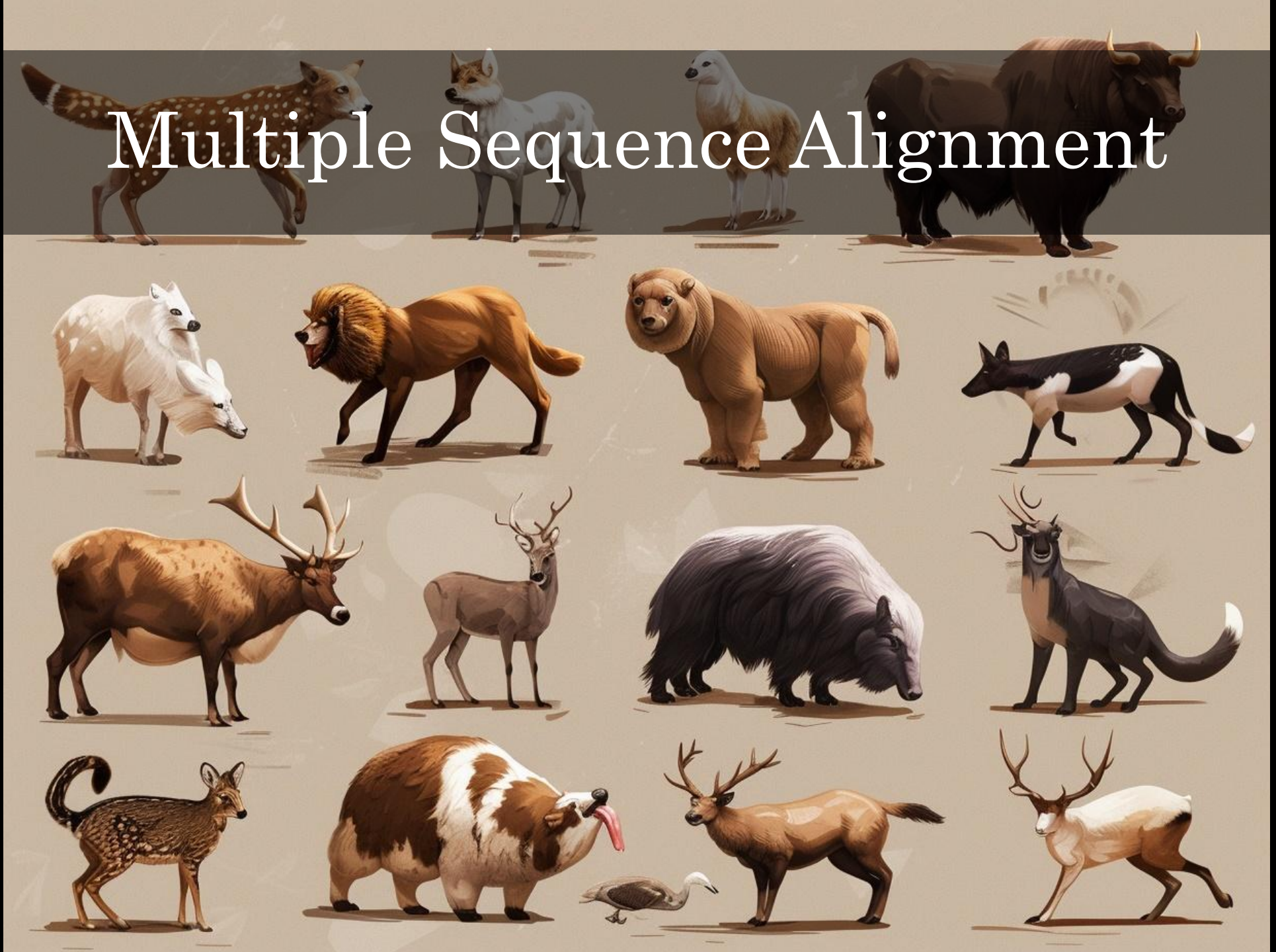
- Quadratic scaling!



Big O complexity

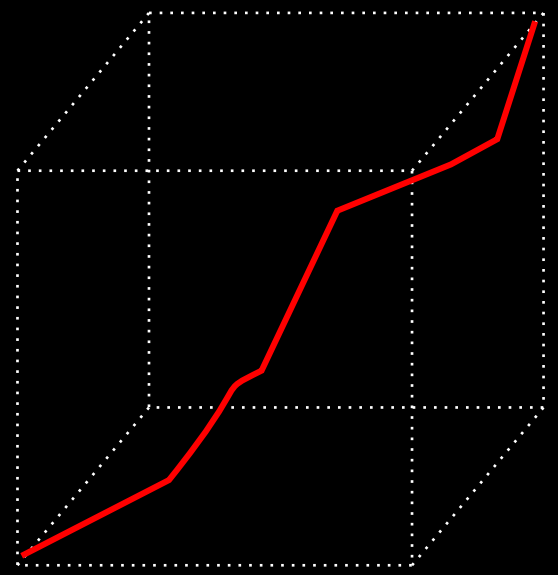


Multiple Sequence Alignment



Multiple Sequence Alignment

- Dynamic programming on k sequences, each of length n requires construction of a k-dimensional matrix with n^k entries



- $= O(n^k)$

- Therefore exponential in the number of sequences!

Scoring MSAs

- In pairwise alignment, we are optimizing the score between two sequences

- When aligning 3 or more sequences, instead optimize the sum of pairs score:

Sequence 1	N
Sequence 2	Q
Sequence 3	Q
Sequence 4	D

SP(N,Q,Q,D) =

S(N,Q)
+ S(N,Q)
+ S(N,D)
+ S(Q,Q)
+ S(Q,D)
+ S(Q,D)

A Key Principle of MSAs

- Aligning everything at once using exact DP = n^k: BAD
- Aligning pairs of sequences using exact DP, then doing something with this information = $\binom{k}{2}(n^2)$: acceptable

An Important Observation

The best alignment between a pair of sequences may not appear in the optimal multiple alignment

ANCDE vs ARCYE

$\hat{a}_{kl}$

a_{kl}

Optimal pairwise alignment

ANCDE
ARCYE

A-NCDE
AR-CYE

Sequences in optimal **multiple** alignment

More sequences, more information

ARNCDE
ARNCYE
…

And

- The score of the optimal multiple alignment $S(a)$ can be <u>no greater</u> than the sum of optimal pairwise alignments $S(\hat{a}^{kl})$

$$\sum_{k<l} S(a^{kl}) \leq \sum_{k<l} S(\hat{a}^{kl})$$

- In general, the multiple alignment score will be less than the sum of all pairwise alignments

But <u>how much less</u>???

MSA (Carrillo and Lipman, 1988)

- If we can establish a lower bound σ on the multiple alignment score, then we constrain each $S(a^{kl})$:

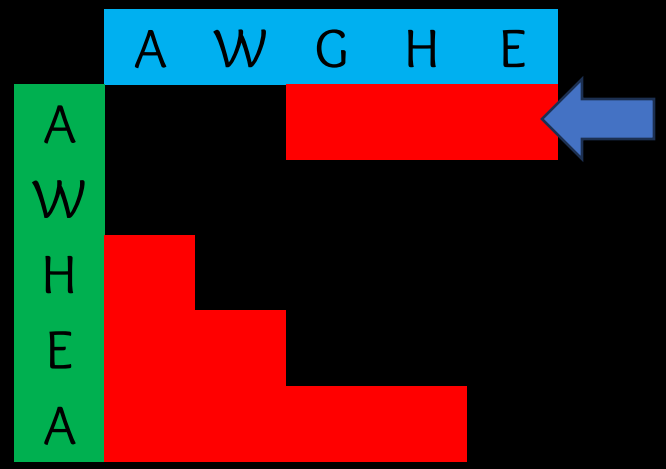
$$S(\hat{a}^{kl}) - S(a^{kl}) \leq \sum_{k'<l'} S(\hat{a}^{k'l'}) - \sigma$$

Remember: sum of all best possible pairwise alignments!

σ high: $S(a^{kl})$ must be close to $S(\hat{a}^{kl})$
Sets a bound on "how much less"

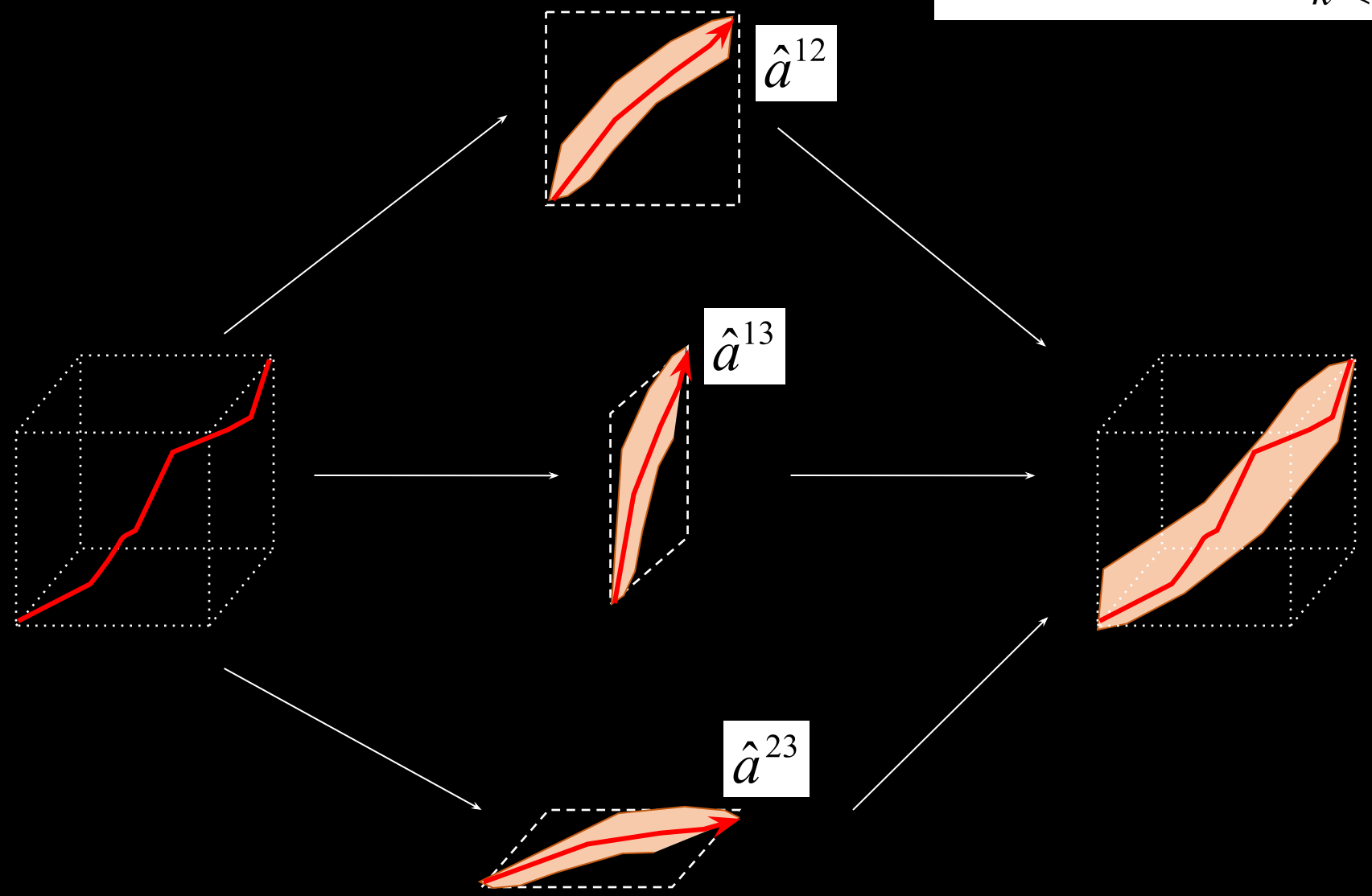
The Consequence of σ

- We can compute $S(\hat{a}^{k'l'})$ for each pair of sequences, and fill the DP grid
- Any cell of the DP grid that gives $S(a^{kl})$ less than σ can be discarded



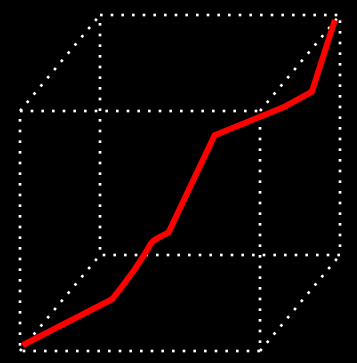
Unacceptable paths: score is too low!

Constrain each pairwise alignment to score no less than $\sigma + S(\hat{a}^{kl}) - \sum_{k'<l'} S(\hat{a}^{k'l'})$



The Last Step

- Multidimensional dynamic programming, restricted to "acceptable" band



- Still $O(n^k)$, but hopefully faster!

So we need all optimal pairwise alignments (again, way cheaper than naïve MSA)

We also need σ. How can we find it?

σ too large: we don't effectively constrain the search space!

σ too small: we may not find an optimal alignment!

Summary

- Dynamic programming allows the calculation of optimal pairwise alignments (for a given scoring scheme!)
- As soon as we go from 2 to >2 sequences, the exponential time complexity of the algorithm makes it impractical
- Need heuristics!